

## Session 3 — How a telescope actually works

*Practical Astronomy Intensive*

Name \_\_\_\_\_

Date \_\_\_\_\_

9 questions      20 minutes

Classroom check with the instruments still on the table. The '600x!' box claim is attacked from three sides — aperture, magnification and the diffraction limit — because it is the misconception this session exists to kill, and it does not die to a single question.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

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- 1 A student has ₹40,000 and asks which is the best telescope design. What is the right answer?
    - a There is no single best — refractor, reflector and catadioptric trade cost, size and image quality, and the best is the one you will actually carry outside
    - b The apochromatic refractor — it is the most expensive design, so it must be the better telescope
    - c Any of them, because optical design is the one specification that decides what a telescope shows
    - d A refractor, because a reflector's secondary mirror and spider block the middle of the image and punch a hole in it
  
  - 2 Two telescopes, both at 100×: a 200 mm Dobsonian and a 100 mm refractor. What does the Dobsonian give you that the refractor cannot?
    - a Fainter stars and finer detail — four times the light grasp, since it climbs as the square of the diameter, and twice the resolving power
    - b Nothing, at the same magnification — the two are showing the same image, since magnification decides how much you can see
    - c Nothing that a dark enough site would not give the 100 mm refractor as well
    - d A bigger image — the larger telescope's job is mainly to make things larger
  
  - 3 A 60 mm department-store refractor has '600× POWER!' printed in yellow across the box. In two or three sentences, explain to the shopper why that number is close to worthless — and say what number they should have looked at instead.

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- 4 Your Dobsonian has a focal length of 1200 mm. You fit a 6 mm eyepiece and then a 25 mm eyepiece. What powers do you get, and which will show more on Jupiter tonight?
- a 200× and 48× — and which shows more depends on the night, since past the aperture's useful ceiling and on unsteady air the higher power just serves up a bigger blur
  - b 200× and 48× — the 6 mm always shows more detail, because more power always means more detail
  - c The powers cannot be worked out from this — magnification is a property of the telescope, and the eyepiece is a minor accessory
  - d 7200× and 30000× — you multiply the two focal lengths
- 5 The 60 mm box scope from the shop really does come with an eyepiece that yields 600×. If you fit it and point at Saturn, what will you actually see?
- a Saturn much larger and with far more detail than at 60× — that is what the extra power is for
  - b A dim, shaking, badly blurred smear — 600× is far past what 60 mm can usefully deliver, so it enlarges without adding any information
  - c A large, sharp Saturn, though a dim one — magnification costs brightness but never detail
  - d Nothing at all — the eyepiece is the wrong size for the scope and will not come to focus
- 6 A student who does daytime photography says 'an f/5 telescope is faster, so stars will look brighter through it than through an f/10.' Is she right?
- a No — the f/10 is brighter, since a longer focal length collects light over a longer path
  - b Yes — a fast f-ratio gathers light more efficiently, so everything looks brighter through it
  - c Yes, and more generally f/5 is simply the better telescope — fast optics are better optics
  - d No — visually, aperture and magnification set the brightness, and the f-ratio just follows from them; her camera intuition does not carry across to the eyepiece
- 7 A 200 mm Dobsonian at 20× gives an exit pupil of 10 mm. A student, aged 55, argues this is ideal because a bigger exit pupil always means more light into the eye. Where does the argument fail?
- a It fails because a 10 mm exit pupil causes blackouts and kidney-beaning, which is a fault in the eyepiece design
  - b His night pupil opens to about 5 mm, so half the beam lands on his iris and is thrown away — the extra aperture is wasted at that power
  - c It does not fail — a bigger exit pupil is always better, since more of the beam reaches the eye
  - d It fails only because he has not accounted for eye relief, which is purely a comfort matter anyway

- 8 A tight double star will not split in a 60 mm refractor. A student swaps to a shorter eyepiece, then adds a Barlow, and is baffled that the blur only gets larger. Why?
- a He has not gone far enough — with enough magnification any double will eventually split
  - b The tiny disc and rings around each star are a flaw in that cheap objective — a better 60 mm would resolve it
  - c The refractor's focal length is too short — a longer 60 mm scope would resolve finer
  - d Diffraction sets a hard floor — the Airy discs of 60 mm simply overlap at that separation, and power only enlarges the overlapping blur
- 9 A cold front has just passed. The night is crystal-clear, sparkling, with the stars twinkling hard and a brisk wind. A student declares it perfect for Jupiter. What should you tell him?
- a He is right, and he should carry the scope straight out and observe immediately to make the most of it
  - b Transparency is excellent but the seeing is probably terrible — that hard twinkle is the same air boiling Jupiter's disc into mush; a hazy still night usually beats it for planets
  - c He is right — a clear night with no haze is by definition good for everything, planets included
  - d He is right about the sky, and if Jupiter still shimmers into mush the telescope's optics are the problem